

Explainable AI - Banana Problem Unit 3 Report

Student Details

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College: PES University

Section: F

Subject: Explainable AI

Unit: 3 - Visual Explanations for Banana Classification

1. Objective

For odd SRNs, the assignment requires Task 1 and Task 2 only. A pretrained ResNet18 model trained on ImageNet was used to study banana class explanations through vanilla gradient saliency and activation maximization.

2. Input Image and Prediction

Input image:

/Users/vivekgowdas/Downloads/realistic-3d-cartoon-banana-peeling-icon_23-2152015160.jpg.avif

Predicted probability for banana class: 0.4283

3. Top Model Predictions

banana: 0.4283

cup: 0.1201

hair slide: 0.0585

lipstick: 0.0344

nipple: 0.0297

4. Task 1 - Vanilla Gradient Saliency

Target logit for banana class: 10.9276

Interpretation: The saliency map mainly emphasizes object boundaries and contour-like regions. Some noisy background response is still visible, which is expected with vanilla gradients.

5. Task 2 - Activation Maximization

Final optimized banana logit: 86.8431

Interpretation: Activation maximization produces a smooth yellow curved pattern instead of a realistic banana photo. This shows the pretrained CNN relies strongly on banana-like textures and curved shape cues.

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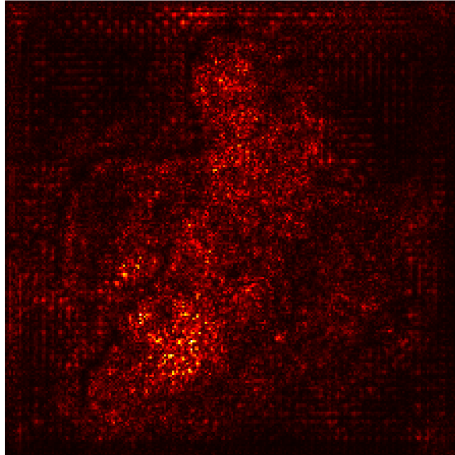
6. Visualizations

6.1 Vanilla Gradient Saliency Map

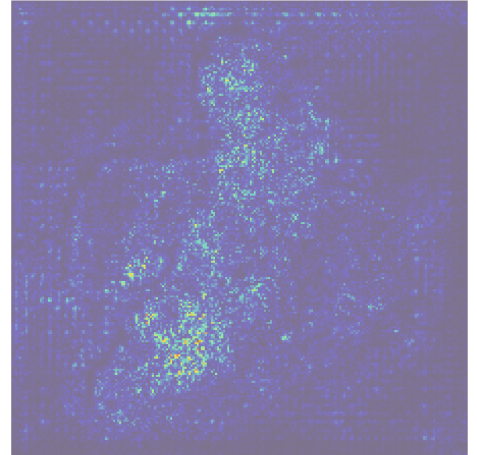
Original Banana Image



Vanilla Gradient Saliency

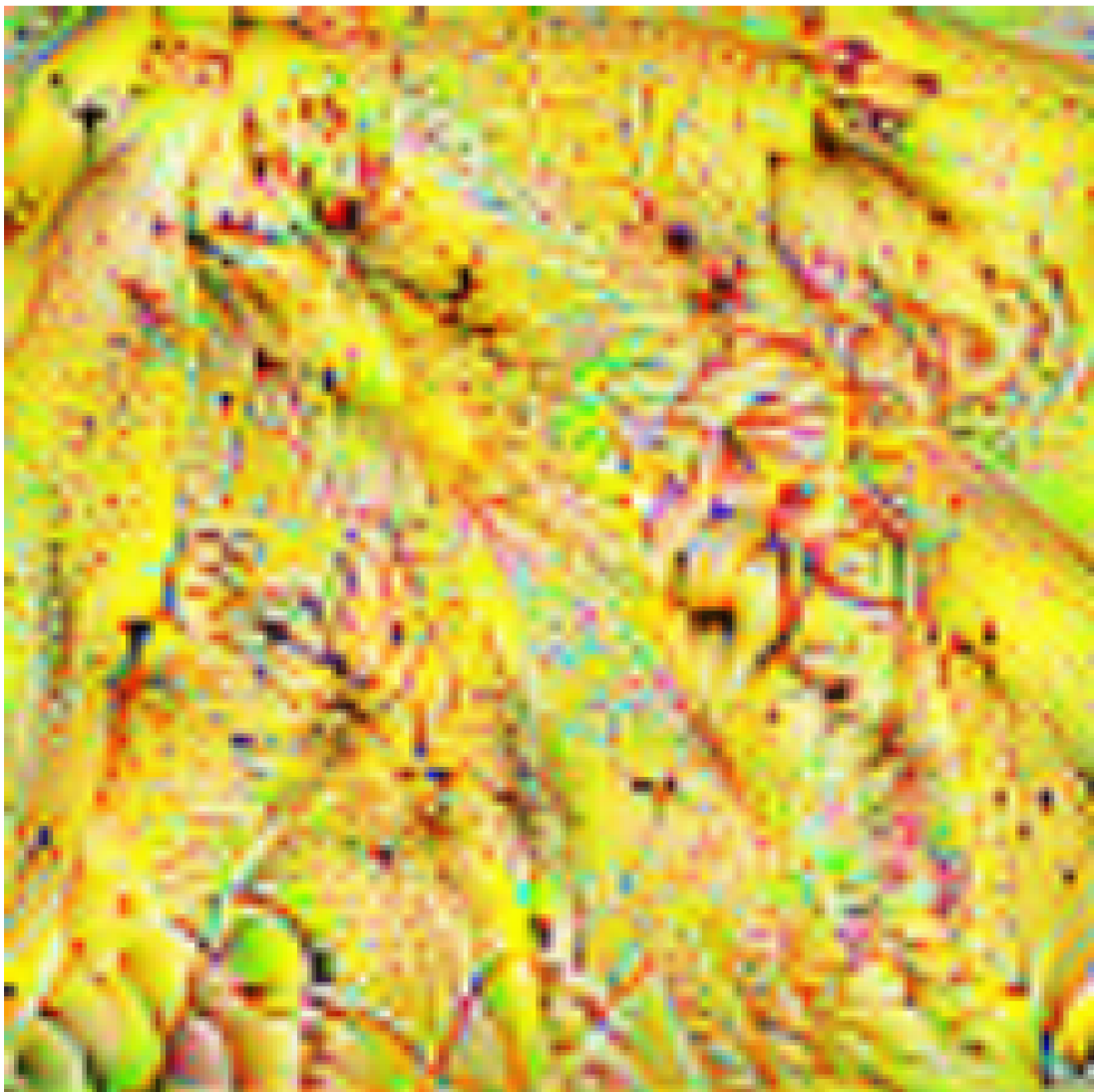


Overlay



6.2 Activation Maximization

Activation Maximization for Banana Class



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7. Conclusion

Vanilla gradients highlighted edge-heavy regions and were visibly noisy, which matches the expected behavior of gradient-based saliency. Activation maximization produced a yellow curved texture pattern rather than a clean object, indicating that the CNN encodes banana evidence through a mixture of color and curved-shape cues.

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